EXP: 6 SEQUENCE DIAGRAM

AIM: To design and develop a SEQUENCE DIAGRAM for the system "PRECISION DISASTER RECOVERY (PDR-SYNC)" to represent the interaction between system components in a time-ordered manner.

THEORY:

A SEQUENCE DIAGRAM is a UML behavioral diagram that shows how objects interact in a particular sequence of events. It illustrates the flow of messages between different components of a system over time. It is mainly used to:

- Understand system workflow
- Model dynamic behavior
- Represent real-time interactions
- Support system design and debugging

ELEMENTS OF SEQUENCE DIAGRAM:

ACTOR: Represents an external entity interacting with the system.
Example: Field Responder

OBJECT / LIFELINE: Represents system components involved in interaction.
Example: Report Controller, Normalization Engine

MESSAGE: Represents communication between objects.
Example: Submit Report, Sync Data

ACTIVATION BAR: Represents the time an object is active

RETURN MESSAGE: Shows response from one object to another

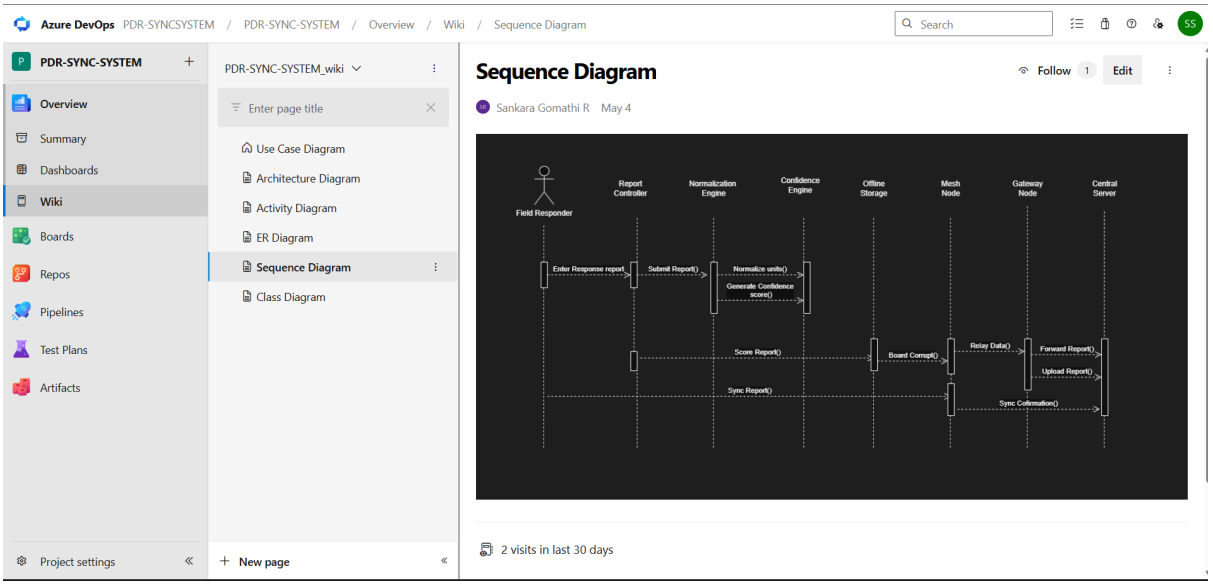
NOTATIONS USED:

NOTATION	DESCRIPTION
Stick Figure	Actor
Rectangle with dashed line	Lifeline
Arrow	Message
Thin Rectangle	Activation

Dashed Arrow Return Message

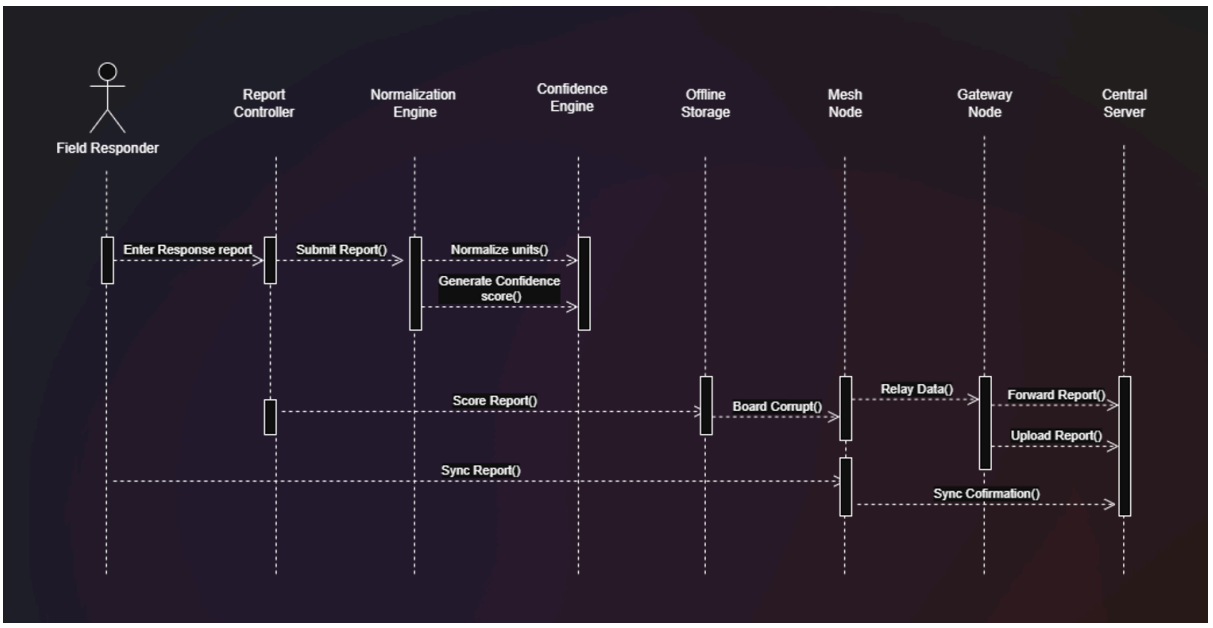
PROCEDURE:

1. Identify the scenario to model (report submission and synchronization)
2. Identify actors and system components involved
3. Arrange components in sequence from left to right
4. Define the flow of messages between components
5. Represent processing steps using activation bars
6. Ensure correct order of execution
7. Validate the diagram with system functionality



STEPS TO CREATE SEQUENCE DIAGRAM IN AZURE:

1. Login to Azure Portal
2. Open Azure DevOps Project
3. Navigate to Repos or Wiki
4. Use diagram tools such as draw.io or Lucidchart
5. Create lifelines for system components
6. Add messages between components
7. Arrange interactions in proper sequence
8. Export the diagram as an image
9. Upload it into Azure DevOps



EXPLANATION OF SEQUENCE:

The sequence diagram represents the workflow of submitting and synchronizing a disaster report.

1. The FIELD RESPONDER enters a response report
2. The REPORT CONTROLLER receives and processes the report
3. The NORMALIZATION ENGINE converts units into standard format
4. The CONFIDENCE ENGINE generates a confidence score
5. The processed report is stored in OFFLINE STORAGE
6. The system prepares the report for synchronization
7. The MESH NODE relays the data across nearby devices
8. The GATEWAY NODE forwards the report to the central system
9. The CENTRAL SERVER uploads and stores the report
10. A SYNC CONFIRMATION is sent back to the system

This sequence ensures:

- Offline-first operation
- Reliable data transmission
- Real-time synchronization when network is available

RESULT:

Thus, the SEQUENCE DIAGRAM for the system “PRECISION DISASTER RECOVERY (PDR-SYNC)” was successfully designed, representing the interaction between system components in a sequential manner.